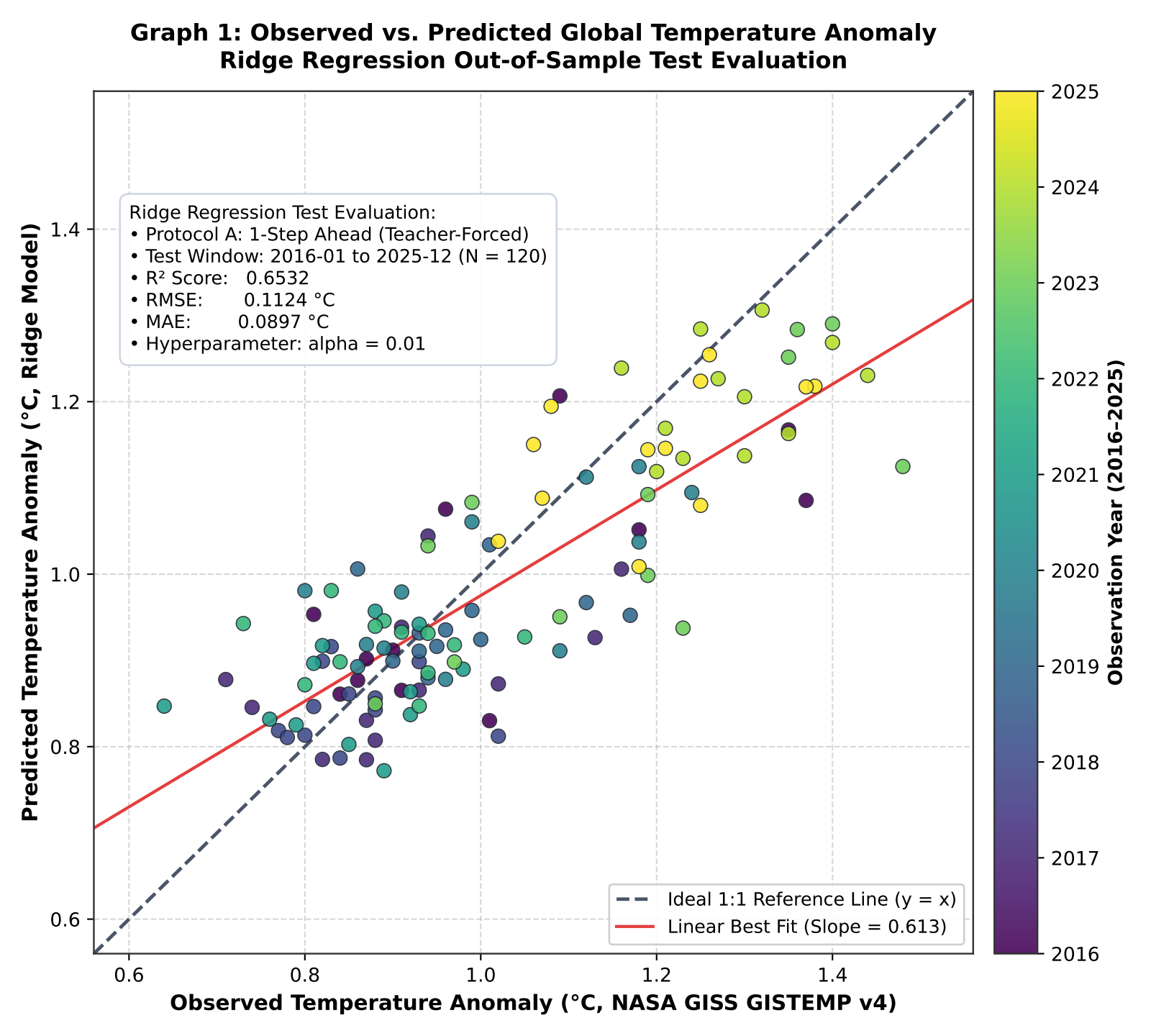
Graph 1: Observed vs. Predicted Global Temperature Anomaly
Ridge Regression Out-of-Sample Test Evaluation
Ridge Regression Test Evaluation:
• Protocol A: 1-Step Ahead (Teacher-Forced)
• Test Window: 2016-01 to 2025-12 (N = 120)
• R² Score: 0.6532
• RMSE: 0.1124 °C
• MAE: 0.0897 °C
• Hyperparameter: alpha = 0.01
Ideal 1:1 Reference Line (y = x)
Linear Best Fit (Slope = 0.613)
Predicted Temperature Anomaly (°C, Ridge Model)
Observed Temperature Anomaly (°C, NASA GISS GISTEMP v4)
Observation Year (2016–2025)
0.6
0.8
1.0
1.2
1.4
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025